Athiya Deviyani

PhD student, Language Technologies Institute, Carnegie Mellon University adeviyan@cs.cmu.edu athiyadeviyani.github.io scholar github linkedin

RESEARCH INTERESTS

Principled and efficient measurement of AI systems using tools from measurement theory, causal inference, and statistics

EDUCATION

Carnegie Mellon University, Pittsburgh, PA, United States Aug 2023 - May 2029 (expected)
Ph.D. Language and Information Technologies, School of Computer Science
- Advisor: Dr. Fernando Diaz; medical leave of absence, academic year 2025-2026
- Relevant courses: Probability and Mathematical Statistics

Carnegie Mellon University, Pittsburgh, PA, United States Aug 2021 - May 2023
M.S. Artificial Intelligence and Innovation
- Advisors: Dr. Michael Shamos, Dr. Alan W. Black, Dr. Maarten Sap
- Relevant courses: Advanced Deep Learning, Advanced NLP, ML with Large Datasets, Computational Ethics for NLP, Deep Reinforcement Learning

The University of Edinburgh, Edinburgh, United Kingdom Sep 2017 - May 2021
B.Sc. (Hons) Artificial Intelligence and Computer Science, first class honours
- Academic award: 2021 Class Prize for Artificial Intelligence and Computer Science (top of class, pdf)
- Honours thesis: *Assessing Dataset Bias in Computer Vision* (outstanding classification, blind double marking)
- Research advisors: Dr. Ajitha Rajan, Dr. Hakan Bilen, Dr. Steven R. Wilson
- Relevant courses: Machine Learning, Foundations of NLP, Automatic Speech Recognition

PUBLICATIONS

Efficient Best-of-N Policy Evaluation for Inference-Time Alignment
Jonas Schweisthal, Yuxin Wang, **Athiya Deviyani**, Stefan Feuerriegel, Dennis Frauen
Preprint, 2026 pdf

Causal Methods for LLM Development and Evaluation
Dennis Frauen, Marie Brockschmidt, Konstantin Hess, Haorui Ma, Yuchen Ma, Abdurahman Maarouf, Maresa Schröder, Jonas Schweisthal, Yuxin Wang, **Athiya Deviyani**, Sonali Parbhoo, Rahul G. Krishnan, Stefan Feuerriegel
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2026, Blue Sky Ideas Track pdf

Nonparametric LLM Evaluation from Preference Data
Dennis Frauen, **Athiya Deviyani**, Mihaela van der Schaar, Stefan Feuerriegel
International Conference on Machine Learning (ICML) 2026 pdf

Quantifying the Statistical Effect of Rubric Modifications on Human-Autorater Agreement
Jessica Huynh, Alfredo Gomez, **Athiya Deviyani**, Renee Shelby, Jeffrey P. Bigham, Fernando Diaz
Preprint, 2026 pdf

Contextual Metric Meta-Evaluation by Measuring Local Metric Accuracy
Athiya Deviyani, Fernando Diaz
Findings of the Association for Computational Linguistics: NAACL 2025 pdf poster

Putting the Person in Personalization: Towards Virtual User Identification via Contrastive Session Pair Discrimination
Athiya Deviyani, Cheng Ju, Kevin Zielnicki, Patrick Glynn
Preprint, 2025 pdf

Men Have Feelings Too: Debiasing Sentiment Analyzers using Sequence Generative Adversarial Networks
Athiya Deviyani, Mehak Malik, Haris Widjaja, Dan Hoskins
AI for Social Good Workshop, AAAI 2023 (oral presentation) pdf code

Text Normalization for Speech Systems for All Languages
Athiya Deviyani, Alan W. Black
Speech for Social Good Workshop, INTERSPEECH 2022 (oral presentation) pdf

Assessing Dataset Bias in Computer Vision
Athiya Deviyani
The University of Edinburgh Outstanding Undergraduate Dissertations 2021 pdf

PATENTS

Virtual User Identification Using Proxy-Labeled Session Pair Discrimination
Athiya Deviyani, et al. (Netflix)
U.S. Patent Application, published Jul 2026

ACADEMIC RESEARCH EXPERIENCE

Language Technologies Institute, Carnegie Mellon University, Pittsburgh, PA Dec 2021 - May 2023
Graduate Research Assistant
- Built a tool for generalizable text normalization for speech synthesis under Dr. Alan W. Black, using the Flite synthesizer and a Python backend (Speech for Social Good Workshop, INTERSPEECH 2022)
- Audited text-to-image diffusion models (DALL·E 2, Stable Diffusion) for gender, race, and age-related social biases under Dr. Maarten Sap, via controlled prompt generation and qualitative and quantitative analysis of generated images using zero-shot CLIP classification; part of a collaboration with AI2 to document the LAION-5B dataset used to train Stable Diffusion

SMASH Lab, The University of Edinburgh, Edinburgh, United Kingdom Jun 2020 - Mar 2021
Research Assistant
- Worked with the Social Media Analysis and Support for Humanity (SMASH) group on using NLP tools (Gensim, FastText, spaCy) and ML classifiers (scikit-learn) to expand the LIWC psycholinguistic lexicons for improved coverage of online slang in social media, using the UrbanDictionary corpus

PROFESSIONAL EXPERIENCE

Netflix, Algorithmic Identity Team May 2024 – 2025
Machine Learning Research Intern, then Part-time Researcher
- Developed a contrastive session-pair model that identifies the individual viewers sharing a single account, trained on proxy labels derived from behavioral signals rather than ground-truth viewer identities
- Deliverables: research paper preprint and U.S. Patent Application
- Presented a poster at the 2024 Netflix Workshop on Personalization, Recommendation and Search (PRS)

Apple, Seattle, WA, United States May 2023 - Aug 2023
Machine Learning Research Intern, Information Intelligence (Search Quality)
- Implemented and evaluated the benefits and drawbacks of data augmentation and parameter-efficient LLM fine-tuning techniques (LoRA, prefix-tuning, prompt-tuning) and SentenceBERT fine-tuning for neural query deduplication, for applications such as Siri search and web answers

Apple, Seattle, WA, United States May 2022 - Aug 2022
Machine Learning Research Intern, Information Intelligence (Search Quality)
- Implemented and evaluated a GraphSAGE graph neural network (R@3 of 0.147) using the Deep Graph Library and PyTorch to suggest related questions for applications such as Siri search and web answers
- Experimented with the effect of diversity penalties in beam search on the diversity and perplexity of query suggestions using distributed training

Goldman Sachs, London, United Kingdom Jun 2021 - Aug 2021
Summer Analyst
- Designed and implemented a data processing pipeline for capital reporting and analysis

Google, London, United Kingdom Jun 2020 - Sep 2020
Software Engineering Intern
- Implemented an algorithm that selects a user's most interesting visits within a selected time frame for the Timeline feature on Google Maps

Google, Zurich, Switzerland Jul 2019 - Sep 2019
Software Engineering Intern
- Built a Channel Shop Tab for creators end to end on the YouTube Creator Commerce team (Growth and User Engagement)

TEACHING EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA, United States Aug 2021 - Dec 2023
Graduate Teaching Assistant · office hours, lectures and recitations, assignment design and grading
- 10-703 Deep Reinforcement Learning (Dr. Katerina Fragkiadaki, Fall 2023)
  - Delivered a lecture on Monte Carlo, temporal difference learning, Q-learning, deep Q-learning, Monte Carlo tree search, REINFORCE, and actor-critic (video)
- 10-707 Advanced Deep Learning (Dr. Ruslan Salakhutdinov, Spring 2023)
- 11-830 Computational Ethics for NLP (Dr. Maarten Sap and Dr. Emma Strubell, Spring 2023)
- 10-617 Intermediate Deep Learning (Dr. Ruslan Salakhutdinov, Fall 2022)
  - Delivered a lecture on generative adversarial networks (GANs) and graph convolutional networks (GCNs) (video)
- 11-737 Multilingual Natural Language Processing (Dr. Graham Neubig, Dr. Alan W. Black, and Dr. Shinji Watanabe, Spring 2022)
- 11-631 Data Science Seminar (Dr. Eric H. Nyberg and Dr. Lori Levin, Fall 2021)

The University of Edinburgh, Edinburgh, United Kingdom Sep 2019 - Jun 2020
Teaching Assistant · tutorial design and delivery, lab demonstrations, and marking
- Introductory Applied Machine Learning (INFR10069, INFR11152; Nigel Goddard), Foundations of Data Science (INFR08030; David Sterratt), Cognitive Science (INFR08020; Matthias Hennig), Software Testing (INFR10057; Ajitha Rajan)

HONORS AND AWARDS

Class Prize for Artificial Intelligence and Computer Science (highest overall GPA), The University of Edinburgh 2021
Outstanding Informatics Undergraduate Thesis, The University of Edinburgh 2021
Best Health Hack, BigRed//Hacks 2021, Cornell University 2021
1st place, TransferWise challenge, HackCambridge 4D 2019
2nd place overall and 3rd place Amazon AWS challenge, OxfordHack 2018

VOLUNTEERING AND SERVICE

Grace Hopper Celebration representative, Carnegie Mellon School of Computer Science 2022
Machine Learning Instructor, Bangkit Academy 2020-2021
Data Science Instructor and Curriculum Organizer, Generation Girl 2020-2021
Python and Web Development Instructor, CodeFirstGirls 2020-2021

SELECTED PROJECTS

Multimodal Emotion Recognition in Conversation with A. Khan, P. Varshney, N. Skarphedinsson pdf poster 2022
Causal Reasoning through Conceptual Explanation Generation with M. Malik, P. Varshney pdf poster 2022
(Visual) Attention is All You Need: Attention-Based Image Super-Resolution with U. Singh pdf code 2022
IndoHateSpeech: Multi-label Hate Speech Detection on Informal Indonesian Corpora with H. Widjaja pdf code 2022
Towards an Unbiased Language Model for Hate Speech Classification pdf code 2022
Gender Obfuscation Methods in Subreddit Classification pdf code 2022
How Real is *Real*: Evaluating the Robustness of Real-World Super Resolution with E. S. Hoplamaz, A. S. Paul pdf code 2021
Informatics Large Practical: Stateless and Stateful Drone Strategies pdf code 2020

SKILLS AND INTERESTS

TECHNICAL Python (PyTorch, TensorFlow), Java, Haskell, SQL, JavaScript, C++, Isabelle, MATLAB
LANGUAGES Indonesian, English, Malay, Mandarin
INTERESTS Piano and violin (ABRSM Grade 5), singing, dance, guitar, Muay Thai, kickboxing